SCoNE: Selective Context-aware Neuron Editing for Robust Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) is highly sensitive to retrieval noise: when retrieved documents mix informative and irrelevant context, LLMs are easily distracted, leading to hallucinations. To overcome this, we propose SCoNE (Selective Context-aware Neuron Editing), a training-free model editing approach that improves retrieval noise robustness by selectively strengthening context-aware FFN neurons that are identified by both high attribution and high cross-input variability. SCoNE requires only a small number of mining samples, no fine-tuning, and no inference-time overhead. Across various knowledge-intensive question-answering benchmarks and two LLM backbones, SCoNE consistently outperforms competitive baseline methods. Our code is available at https://github.com/HYU-ARK-Lab/SCoNE.

1 Introduction

While Large Language Models (LLMs) have achieved remarkable success, they remain prone to hallucinations in knowledge-intensive tasks (Wang and Yu, 2025; Huang et al., 2025). Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) mitigates this by grounding outputs in externally retrieved evidence. However, the effectiveness of RAG is heavily dependent on the quality of retrieved documents, which retrieval systems cannot always guarantee. In realistic settings, retrievers return a mixture of relevant, partially relevant, and irrelevant documents for the same query, and LLMs are known to be easily distracted by such *retrieval noise*, often degrading rather than improving their answers (Yoran et al., 2024; Shi et al., 2023).

Existing approaches to retrieval noise robustness span several paradigms, including prompt engineering, retrieved-context refinement and fine-tuning. While solutions such as introducing additional modules (*e.g.,* reranker, compressor) are flexible, they introduce additional components into the RAG pipeline, which can lead to cascading errors across each stage (Asai et al., 2024; Yoran et al., 2024) and substantial inference-time latency (An et al., 2025). In contrast, directly fine-tuning the generator to be robust against retrieval noise avoids such pipeline overhead and has therefore emerged as a promising direction (Yoran et al., 2024; Wu et al., 2025). However, fine-tuning-based methods inherit the well-known drawbacks of gradient-based adaptation: catastrophic forgetting, substantial compute requirements, and the need for carefully curated training data. A natural question arises: *can retrieval noise robustness be achieved without retraining the model?*

Model editing offers a promising alternative paradigm. By directly modifying a small number of parameters, editing methods provide fine-grained control over model behavior without the cost of fine-tuning. However, existing model editing methods fundamentally assume that the target knowledge to be edited is known in advance (Meng et al., 2022, 2023). This assumption does not hold in Retrieval-Augmented Generation (RAG), where retrieved contexts are inherently open-ended and dynamically vary across queries. In RAG, the model cannot anticipate which facts will appear in the retrieved documents at inference time. Hence, rather than fixing a specific parametric knowledge, RAG requires an adaptive approach to constrain the model's behavior in response to whatever context is retrieved at inference, regardless of its specific content. This challenge is further complicated by the realistic nature of retrieval itself; even for a single query, some documents directly support the answer, others are partially relevant, and others are entirely irrelevant. For this, a method that not only makes the model responsive to context, but selectively responsive to a context (*i.e.,* engaging with

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informative documents while remaining unaffected by noisy ones) is necessarily required.

Building on this insight, we propose **SCoNE** (**S**elective **Co**ntext-aware **N**euron **E**diting) for RAG, which is a model editing method to enhance robustness for retrieval noise. SCoNE identifies *selectively context-aware neurons* by jointly requiring high attribution and high cross-input variability, and strengthens them at inference time. Our method requires only 100 mining samples from a single dataset (HotpotQA), no fine-tuning, and no inference-time overhead beyond standard RAG. Across various benchmarks, SCoNE consistently outperforms strong competitive baselines, demonstrating that lightweight editing, when guided by the right neuron selection criterion, can match or exceed the effectiveness of heavyweight fine-tuning pipelines. Similarly, Shi et al. (2024a) adopt a knowledge-agnostic approach but identify context-aware neurons based on attribution strength alone. While this criterion is effective in their single-context setting, RAG presents multiple retrieved documents containing both informative and distracting evidence simultaneously. Here, a neuron may receive high attribution simply because it responds broadly to any retrieved content, potentially reflecting sensitivity to surface-level patterns rather than informative evidence. Thus, attribution strength alone is insufficient for identifying neurons that specifically mediate useful evidence utilization in noisy retrieval settings. Our redefinition of context-aware neurons addresses this gap by complementing attribution strength with cross-input variability, making the criterion more suitable for RAG's heterogeneous, multi-document setting.

2 Method

Problem Setup. Consider a neuron mining dataset of size N, sampled from the training split of HotpotQA (Yang et al., 2018), $\mathcal{D} = \{(q, \mathcal{C}, a)_t\}_{t=1}^{N}$, where the t-th instance consists of a query q_t, its ground-truth answer a_t, and an associated context set $\mathcal{C}_t = \{c_m^g\}_{m=1}^{M} \cup \{c_k^d\}_{k=1}^{K}$. Here, $\{c_m^g\}_{m=1}^{M}$ denote the gold contexts that directly support a_t, while $\{c_k^d\}_{k=1}^{K}$ are distractor contexts that do not entail the answer. Our goal is to characterize, for each instance, how individual FFN neurons engage with such mixed evidence during inference. To this end, we define two complementary measures, *attribution* and *variability*, computed over the course of inference.

2.1 Neuron Mining

Attribution. Prior work has shown that factual knowledge is localized in specific FFN neurons (Dai et al., 2022; Geva et al., 2021). We hypothesize that neurons responsible for processing retrieved contextual information also reside in FFNs, and seek to identify those that engage with retrieved evidence during RAG inference. Following Shi et al. (2024a), we estimate the attribution of each FFN neuron n via Integrated Gradients (Sundararajan et al., 2017), but extend their single-context formulation to the multi-context RAG setting where gold and distractor evidence co-occur. For each instance $(q, \mathcal{C}, a)_t \in \mathcal{D}$, we quantify how individual FFN neurons contribute to the model's prediction when the query is presented together with its full context set $\mathcal{C}_t = \{c_m^g\}_{m=1}^{M} \cup \{c_k^d\}_{k=1}^{K}$, which contains both gold and distractor contexts. Specifically, we formulate the attribution score calculation as follows: let $v(q_t)$ denote the activation of neuron n when the model is given the query alone, and let $v(q_t, \mathcal{C}_t)$ denote its activation when the query is augmented with the full context set $\mathcal{C}_t$. The attribution score is then defined as follows:

$$\mathrm{Attr}(n;\, q_t, \mathcal{C}_t) = \big(v(q_t, \mathcal{C}_t) - v(q_t)\big) \times \int_{\alpha=0}^{1} \frac{\partial\, P(a \mid q_t, \mathcal{C}_t, v_\alpha)}{\partial\, v_\alpha}\, d\alpha, \tag{1}$$

where $v_\alpha = v(q_t) + \alpha\big(v(q_t, \mathcal{C}_t) - v(q_t)\big)$ linearly interpolates between the query-only and the query-plus-context activations for $\alpha \in [0, 1]$. In practice, the integral is approximated by a 20-step Riemann sum.

Variability. Attribution alone, however, cannot distinguish between two qualitatively different neuron behaviors: neurons that selectively respond to specific contexts and neurons that activate uniformly regardless of context content. Only the former captures the content-level selectivity required in the heterogeneous multi-document setting of RAG. To capture this context selectivity, we measure how a neuron's attribution varies across different query-context instances. Concretely, let $\mathrm{Attr}^{(t)}(n_j^l)$ denote the attribution of the j-th intermediate neuron in the l-th FFN layer for the t-th instance $(q, \mathcal{C}, a)_t \in \mathcal{D}$. The variability score is defined as the deviation of the current attribution from its running average over the preceding W instances:

	NQ	ASQA	SCIQ	TriviaQA	HQA	TruthfulQA	PopQA	Avg.
Llama-3-8B-Instruct								
RAG (Lewis et al., 2020)	62.81	68.78	54.10	88.65	46.55	4.90	60.17	55.14
RetRobust (Yoran et al., 2024)	62.71	69.62	53.10	88.53	46.39	5.14	60.64	55.16
PA-RAG (Wu et al., 2025)	**68.06**	73.73	56.80	90.18	50.41	3.79	64.19	58.17
CAD (Shi et al., 2024b)	64.12	68.99	48.50	87.81	46.77	3.55	62.21	54.56
IRCAN (Shi et al., 2024a)	64.65	71.41	54.00	89.67	50.11	5.51	63.65	57.00
SCoNE (Ours)	66.44	**73.84**	**57.10**	**90.66**	**52.27**	**6.36**	**65.57**	**58.89**
Qwen-2.5-7B-Instruct								
RAG (Lewis et al., 2020)	62.25	69.09	54.90	86.98	45.54	**6.24**	58.54	54.79
RetRobust (Yoran et al., 2024)	60.73	67.19	53.90	86.83	44.41	5.51	57.48	53.72
PA-RAG (Wu et al., 2025)	60.45	68.35	52.90	87.54	**48.00**	6.00	56.65	54.27
CAD (Shi et al., 2024b)	61.83	69.62	51.50	85.62	43.88	4.04	58.84	53.62
IRCAN (Shi et al., 2024a)	62.25	68.78	54.20	87.28	45.86	6.00	58.82	54.74
SCoNE (Ours)	**63.24**	**69.94**	**56.20**	**88.16**	47.00	5.63	**60.04**	**55.74**

Table 1: Accuracy comparison across QA benchmarks using Llama-3-8B-Instruct (top) and Qwen-2.5-7B-Instruct (bottom). **Bold** scores are best in each dataset, and underlined scores are the second-best results.

$$V^{(t)}(n_j^l) = \left| \mathrm{Attr}^{(t)}(n_j^l) - \frac{1}{W} \sum_{m=t-W}^{t-1} \mathrm{Attr}^{(m)}(n_j^l) \right| \quad (2)$$

where W denotes the number of preceding instances used to compute the running average. We compute this score over a fixed traversal order of the mining set $\mathcal{D}$, so that the sliding window captures local variation in the neuron's attribution across neighboring instances in the traversal.

High-Attribution and Variability Neuron Selection. Neurons with both high attribution and high variability are considered context-aware: they contribute strongly to the current context while responding differently across inputs. Hence, we select the neurons as follows: For each sample $(q, \mathcal{C}, a)_t$, we construct $\mathcal{A}^{(t)}$ and $\mathcal{B}^{(t)}$, containing the top-50 neurons ranked by attribution and variability, respectively, restricted to neurons with positive attribution scores.[1] Their intersection $\mathcal{C}^{(t)} = \mathcal{A}^{(t)} \cap \mathcal{B}^{(t)}$ forms the locally selected neurons for sample t. We then aggregate the selection frequency of each neuron across all samples and choose the top-k most frequent neurons as the final context-aware neuron set. For scale, we treat each layer-specific FFN dimension as a distinct neuron. Llama-3-8B-Instruct contains $32 \times 14{,}336 = 458{,}752$ such neurons. Each of the top-50 sets $\mathcal{A}^{(t)}$ and $\mathcal{B}^{(t)}$ corresponds to $\approx 0.0109\%$ of all layer-specific FFN neurons. With $k = 5$, the final neuron set therefore contains $\approx 0.0011\%$ of all layer-specific FFN neurons.

[1]Restricting to positive attribution prevents neurons whose variability stems from transitions between negative and near-zero attribution, whose overall contribution remains negligible, from being selected.

2.2 Neuron Enhancement

Once context-aware neurons are identified, we amplify their contribution at inference time to better leverage informative retrieved evidence. For each selected neuron n_i^l, we scale its corresponding FFN weight as follows: $\hat{W}(n_i^l) = \alpha \cdot W(n_i^l)$, where α controls the enhancement strength.

3 Experimental Setup

Neuron Mining Dataset. We sample the first 100 instances from the HotpotQA (Yang et al., 2018) training split for neuron mining. We set the number of gold-content $M = 2$, and the distractor context $K = 8$.

Evaluation Dataset. We use the dev splits provided by BERGEN (Rau et al., 2024) for NQ (Kwiatkowski et al., 2019), ASQA (Stelmakh et al., 2022), SCIQ (Welbl et al., 2017), TriviaQA (Joshi et al., 2017), HotpotQA (Yang et al., 2018), TruthfulQA (Lin et al., 2022), PopQA (Mallen et al., 2023). All retrieval documents are sourced from the KILT (Petroni et al., 2021) Wikipedia dump[2], and we retrieve top-5 documents per question using SPLADE-v3 (Lassance et al., 2024).

Baselines. We compare SCoNE against representative RAG baselines: RAG (Lewis et al., 2020); RetRobust (Yoran et al., 2024) and PA-RAG (Wu et al., 2025) for generator fine-tuning; and CAD (Shi et al., 2024b) and IRCAN (Shi et al., 2024a) for inference-time intervention at the decoding and parameter level, respectively.

[2]https://huggingface.co/datasets/facebook/kilt_wikipedia

	Relevant			Irrelevant		
Llama-3-8B-Instruct	NQ	SCIQ	HQA	NQ	SCIQ	HQA
RAG (Lewis et al., 2020)	78.04	73.61	72.75	4.43	**8.36**	15.16
PA-RAG (Wu et al., 2025)	**85.29**	**78.60**	**81.26**	2.04	5.69	13.43
IRCAN (Shi et al., 2024a)	80.44	73.75	76.88	4.09	7.69	18.02
SCoNE (Ours)	82.00	78.03	79.53	**6.81**	8.03	**19.59**

Table 2: The comparison of accuracy on Relevant and Irrelevant subsets.

Measure	**NQ**	**SCIQ**	**HQA**
Variance / Std.	63.52	54.70	50.25
Mean Absolute Deviation (MAD)	64.61	54.10	50.27
SCoNE	66.44	57.10	52.27

Table 3: The comparison of the proposed variability with order-invariant measures for neuron selection on Llama-3-8B-Instruct.

Implementation Details. Our experiments are performed using the RAG framework provided by BERGEN (Rau et al., 2024), which offers a realistic RAG pipeline. We use Llama-3-8B-Instruct (Llama Team, 2024) and Qwen-2.5-7B-Instruct (Yang et al., 2024) as the generator LLM, and SPLADE-v3 (Lassance et al., 2024) as the retriever. For retrieval, we use the KILT Wikipedia dump[3], preprocessed into non-overlapping 100-word chunks, and retrieve five documents per question. All experiments are conducted on a single NVIDIA H200 GPU. For fair comparison, both IRCAN and our method identify neurons from the same neuron mining dataset $\mathcal{D}$. Details of $\mathcal{D}$ are provided in A.1. We set the enhancement strength $\alpha = 7$, select the top-k context-aware neurons where $k = 5$, and the window size $W = 3$.

4 Results

Main Results. Table 1 reports the main results. Accuracy is measured using the Match score, which checks whether the gold answer appears as a substring of the generated output. SCoNE achieves the best overall performance with Llama-3-8B-Instruct, ranking first on six of seven datasets and improving over RAG by 3.75% on average. Compared to fine-tuning baselines, SCoNE outperforms RetRobust by 3.73% and remains within 0.72% of PA-RAG despite requiring no additional training. Against intervention-based baselines, SCoNE surpasses CAD on all datasets and improves over IRCAN by up to 3.1% on SCIQ using Llama-3-8b-Instruct, suggesting that our variability-based criterion identifies neurons more selectively responsive to retrieved context than attribution alone. A similar trend holds with Qwen-2.5-7B-Instruct as the generator, where SCoNE consistently outperforms IRCAN across most benchmarks. SCoNE achieves the best average accuracy, demonstrating its effectiveness across different generators. This advantage is also preserved under LLM-based evaluation (Appendix A.4). We confirm that SCoNE's improvements stem from neuron selection: randomly selected neurons remain on par with vanilla RAG (Appendix A.5). We further verify that these gains are robust to the choice of mining sample, with accuracy remaining stable across different 100-example samples from HotpotQA (Appendix A.6).

Relevant vs. Irrelevant Context Analysis. To examine whether our method exhibits context-dependent behavior with retrieved contexts, we divide each evaluation set into two subsets: *Relevant*, where at least one retrieved document contains the gold answer, and *Irrelevant*, otherwise. As shown in Table 2, SCoNE consistently improves over RAG and IRCAN on the relevant subset across all datasets. While PA-RAG achieves the highest accuracy on the relevant subset, SCoNE demonstrates stronger robustness under irrelevant contexts, outperforming all baselines, including vanilla RAG, on NQ and HQA. This robustness holds under a controlled noise experiment (Appendix A.12). Notably, SCoNE surpasses IRCAN on both the Relevant and Irrelevant subsets, suggesting that incorporating cross-input variability beyond attribution strength helps identify neurons that selectively engage with informative evidence. This selectivity is reflected in their activation patterns across different retrieved-evidence compositions (Appendix A.11).

Comparison of Variability Measures Our variability measure in Eq. 2 uses an unsigned running residual and thus depends on example order. We compare it with three order-invariant alternatives—variance, standard deviation, and mean absolute deviation (MAD)—computed over the full set of attribution scores for each neuron, irrespective of their traversal order. With all other settings fixed, Table 3 shows that SCoNE consistently outperforms these measures across all three datasets on Llama-3-8B-Instruct. Variance and standard deviation yield identical results because they induce the same neuron ranking and select the same top-5 neurons. Overall, the results suggest that SCoNE's running-residual formulation provides a more effective variability signal for identifying selective

[3]https://huggingface.co/datasets/kilt_wikipedia

Selection	NQ	SCIQ	HotpotQA
Attr-only	64.61	54.10	50.27
Var-only	63.52	54.70	50.25
Attr+Var (SCoNE)	**66.44**	**57.10**	**52.27**

Table 4: The comparison of neuron selection criteria on Llama-3-8B-Instruct.

context-aware neurons.

Ablation on Neuron Selection Criteria To isolate the contribution of cross-input variability beyond attribution alone, we compare three neuron-selection strategies while keeping all other settings identical: Attr-only, Var-only, and Attr+Var (SCoNE). In Table 4 Attr+Var consistently outperforms Attr-only by 1.83, 3.00, and 2.00% on NQ, SCIQ, and HotpotQA, respectively. Var-only is insufficient by itself, whereas its combination with attribution consistently yields the best performance. This demonstrates that variability provides a complementary and substantive signal for neuron identification.

Hyperparameter Analysis. We conduct ablation studies on the enhancement strength α, context window size W, neuron mining dataset size N, and the number of selected neurons k using Llama-3-8B-Instruct. The result is shown in Figure 1.[4] Performance improves monotonically with α, peaking at $\alpha = 7$, and remains stable across small-to-moderate W, N, and k. Overall, while extreme hyperparameter values (*e.g.*, $W = 10$ or $N = 1000$) degrade performance, the default SCoNE configuration denoted as *SCoNE (Ours)* consistently achieves the best or near-best performance, validating our design choices.[5]

5 Related Work

Retrieval Noise Robustness in RAG. Prior work has addressed retrieval noise in Retrieval-Augmented Generation (RAG) through prompt engineering (Zhou et al., 2023), analyses of retrieved-context composition (Cuconasu et al., 2024), retrieved-context refinement, and fine-tuning. Retrieved-context refinement includes reranking and compression (Glass et al., 2022; Xu et al., 2024), with recent approaches further exploring compact clue selection (Zhang et al., 2026a), reinforcement-learning-based evidence extraction (Zhao et al., 2026), and attention-based context compression (Zhang et al., 2026b). Fine-tuning approaches instead adapt the generator itself to improve robustness against retrieval noise. Yoran et al. (2024) train the generator on mixtures of relevant and irrelevant contexts, while Wu et al. (2025) align it via multi-perspective preference optimization. More recently, Wu et al. (2026) incorporate conflict signals into multi-stage learning to improve robustness against conflicting retrieved knowledge.

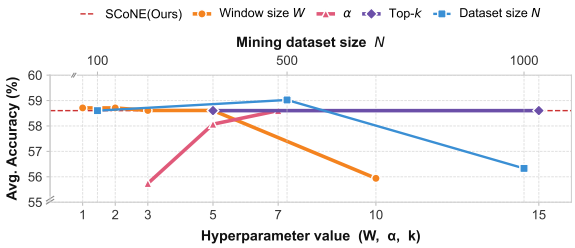

Figure 1: Ablation study on key hyperparameters using Llama-3-8B-Instruct. Results are averaged across NQ, SCIQ, and HotpotQA.

Model Editing for Context Utilization. Model editing modifies model parameters to alter knowledge or behavior without additional training. Methods typically target specific knowledge known in advance (Meng et al., 2022, 2023). Recent studies have extended model-level interventions toward knowledge-agnostic control of contextual knowledge utilization. Shi et al. (2024a) perform neuron-level model editing by identifying and reweighting context-aware neurons based on attribution strength, enabling knowledge-agnostic adaptation to contextual knowledge. SCoNE builds on this knowledge-agnostic, neuron-level perspective for noisy multi-document RAG, where informative and distracting contexts coexist. It therefore complements attribution strength with cross-input variability to identify selectively context-responsive neurons.

6 Conclusion

We present SCoNE, a framework that improves RAG robustness to retrieval noise by selectively enhancing context-aware neurons, identified through attribution strength and cross-input variability. Experiments across various benchmarks show that SCoNE matches or surpasses strong baselines, demonstrating that variability-based neuron mining provides a practical criterion.

[4] More detailed results for each hyperparameter setting are provided in A.7, A.9, A.10.

[5] SCoNE (Ours) corresponds to the default configuration with $\alpha = 7$, $W = 3$, $N = 100$, and $k = 5$.

Limitations

While the selected neurons transfer effectively across diverse benchmarks, several limitations remain. In this work, neuron mining is performed using samples from HotpotQA only, and it remains unclear how the characteristics of the mining dataset influence the selected neuron set and downstream behavior. For example, using more challenging QA datasets may lead to different neuron distributions and transfer properties. In addition, our current setting relies on contexts containing both gold-content and distractor-content. Therefore, it remains unclear how neuron selection would differ under cleaner retrieval settings containing only gold supporting documents. Investigating how mining dataset composition and retrieval conditions affect neuron mining and robustness remains an important direction for future work.

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A Appendix

A.1 Neuron Mining Dataset Setting

For neuron mining, we use the first 100 samples from the HotpotQA training distractor split (Yang et al., 2018). To simulate a realistic RAG setting where multiple retrieved documents are provided as input, we convert each sample into the context-provided prompt format shown in Table 5, where all documents in the sample are directly used as the input context without an additional retriever. The corresponding prompt format without retrieved context is shown in Table 6.

Input Format With Retrieved Context

System: Your task is to extract relevant information from provided documents and to answer questions as briefly as possible.

User:

Background:

Document 1: [title] [sentences]

Document 2: [title] [sentences]

⋮

Document N: [title] [sentences]

Question: [question]

Table 5: Input prompt format with retrieved context, used for attribution score computation.

Input Format Without Retrieved Context

System: Answer the questions as briefly as possible.

User: Question: [question]

Table 6: Input prompt format without retrieved context, used for attribution score computation.

A.2 Baseline Implementation Details

To ensure a consistent comparison, for `PA-RAG`, we use the authors' publicly released Llama-3-8B-Instruct checkpoint and reproduce the method on Qwen-2.5-7B-Instruct using their training recipe. For `RetRobust`, we reproduce results on LLaMA-3-8B-Instruct and Qwen-2.5-7B-Instruct using the authors' training recipe. For `CAD`, we set $\alpha = 0.5$. For `IRCAN`, we use the same backbone as `SCoNE` and select the top 20 neurons by attribution as the candidate pool, with $\alpha = 7$ and $k = 5$ final neurons.

Inference Input Format

System: You are a helpful assistant. Your task is to extract relevant information from provided documents and to answer questions as briefly as possible.

User:

Background:

Document 1: ...

Document 2: ...

⋮

Document 5: ...

Question: [question]

Table 7: Input prompt format used for inference-time RAG.

Method	SCIQ	NQ	HQA
RAG	67.00	57.03	52.14
PA-RAG	60.80	55.66	52.21
CAD	59.40	56.75	47.12
IRCAN	67.80	57.81	**54.70**
SCoNE (Ours)	**67.90**	**58.69**	54.39

Table 8: LLM-based evaluation results on Llama-3-8B-Instruct using GPT-5-mini as the judge.

A.3 Inference-Time RAG Prompt

The prompt format used for inference-time RAG generation is shown in Table 7. The "Background" section consists of the top-5 documents retrieved by the retriever.

A.4 LLM-based Evaluation

We evaluate model outputs using GPT-5-mini as an LLM judge on NQ, SCIQ, and HotpotQA with Llama-3-8B-Instruct. We use the LLM-as-a-judge evaluation protocol provided by Rau et al. (2024). As shown in Table 8, SCoNE achieves the highest average score and the best performance on two of the three datasets.

A.5 Random Neuron Selection

To test whether SCoNE 's gains depend on neuron selection, we select five random neurons and apply the identical enhancement strength on Llama-3-8B-Instruct. As shown in Table 9, random neuron selection yields performance comparable to vanilla RAG across all three datasets. In contrast, SCoNE, using the same strength, achieves the best performance across datasets, suggesting that neuron selection drives the gain.

Method	NQ	SCIQ	HQA	Avg.
Random (seed 42)	62.71	54.10	46.68	54.50
Random (seed 77)	62.53	53.90	46.70	54.38
Random (seed 99)	62.46	54.10	46.48	54.35
Random (seed 512)	62.99	53.80	46.45	54.41
Random (seed 256)	62.81	53.90	46.54	54.42
RAG	62.81	54.10	46.55	54.49
SCoNE	**66.44**	**57.10**	**52.27**	**58.60**

Table 9: Random-neuron control experiment on Llama-3-8B-Instruct. Each random selection uses the same number of neurons $k = 5$ and enhancement strength $\alpha = 7$ as SCoNE.

Mining Set	Accuracy
First 100 (Ours)	52.27
Seed 11	53.75
Seed 55	53.14
Seed 741	52.77
Seed 333	52.64
Seed 512	52.54
Seed 150	52.50
Seed 421	52.48
Seed 107	52.46
Seed 909	52.39
Avg. $\pm$ Std.	52.69 ± 0.44

Table 10: Sensitivity to the choice of neuron-mining examples on HotpotQA using Llama-3-8B-Instruct. The first row uses the initial 100 training examples, while the remaining rows use 100 examples randomly sampled with different seeds.

A.6 Sensitivity to Mining Samples

We show that SCoNE maintains its performance across different sets of 100 examples used for mining. The seed only determines which 100 examples are drawn from the HotpotQA training split. We therefore mined neurons with various random seeds on Llama-3-8B-Instruct and evaluated on HotpotQA. As shown in Table 10, match accuracy remains stable across seeds (52.69 ± 0.44).

A.7 Effect of Enhancement Strength and Number of Selected Neurons

We additionally evaluate different enhancement strengths $\alpha \in \{3, 5, 7\}$ and top-k values $k \in \{5, 15\}$ on NQ, HotpotQA, and SCIQ using Llama-3-8B-Instruct. Larger enhancement strengths generally lead to better performance, with $\alpha = 7$ achieving the strongest overall results. We also observe that $k = 5$ and $k = 15$ yield comparable performance, suggesting that the neurons most important for selective context utilization are already concentrated within a small set of top-ranked neurons.

	NQ		SCIQ		HQA	
α	k=5	k=15	k=5	k=15	k=5	k=15
3	63.80	63.27	53.50	54.00	49.91	49.57
5	65.81	65.67	56.20	58.00	52.20	52.66
7	66.44	66.02	57.10	57.70	52.27	52.09

Table 11: Effect of enhancement strength α and top-k neuron mining.

Pool Size	HQA	SCIQ	NQ
Llama-3-8B-Instruct			
Top-20	**52.46**	**57.90**	**66.55**
Top-50	52.27	57.10	66.44
Top-80	50.25	54.70	63.52
Qwen-2.5-7B-Instruct			
Top-20	46.04	53.90	62.50
Top-50	**47.00**	**56.20**	**63.24**
Top-80	46.04	53.90	62.50

Table 12: Effect of candidate pool size.

A.8 Effect of Candidate Pool Size

To validate the choice of 50 candidate neurons, we vary the size of $A^{(t)}$ and $B^{(t)}$ over $\{20, 50, 80\}$ before taking their intersection. Table 12 reports the results on HotpotQA, SCIQ, and NQ using Llama-3-8B-Instruct and Qwen-2.5-7B-Instruct.

We observe that Top-20 is marginally higher than Top-50 by 0.1-0.8 points on Llama-3-8B-Instruct, but on Qwen-2.5-7B-Instruct, Top-50 is best on all three datasets by 0.7-2.3 points, so Top-50 offers the best overall trade-off. Top-20 and top-80 select the same final neuron set; therefore, they yield identical scores.

A.9 Effect of Context Window Size

We vary the context window size $W \in \{1, 2, 3, 5, 10\}$ used to compute attribution variability. As shown in Figure 2, performance remains stable across small and moderate window sizes ($W = 1, 2, 3, 5$), while a large window ($W = 10$) consistently degrades performance across datasets. We additionally observe that the mined neuron sets are highly similar across different window sizes. In particular, the selected neurons for $W = 1$ and $W = 2$ are identical, and those for $W = 3$ and $W = 5$ are also identical. Even for $W = 10$, the selected neuron set differs from the other settings by at most one neuron. Despite these highly similar neuron sets, performance differences remain relatively small for practical window sizes, with degradation mainly observed at $W = 10$.

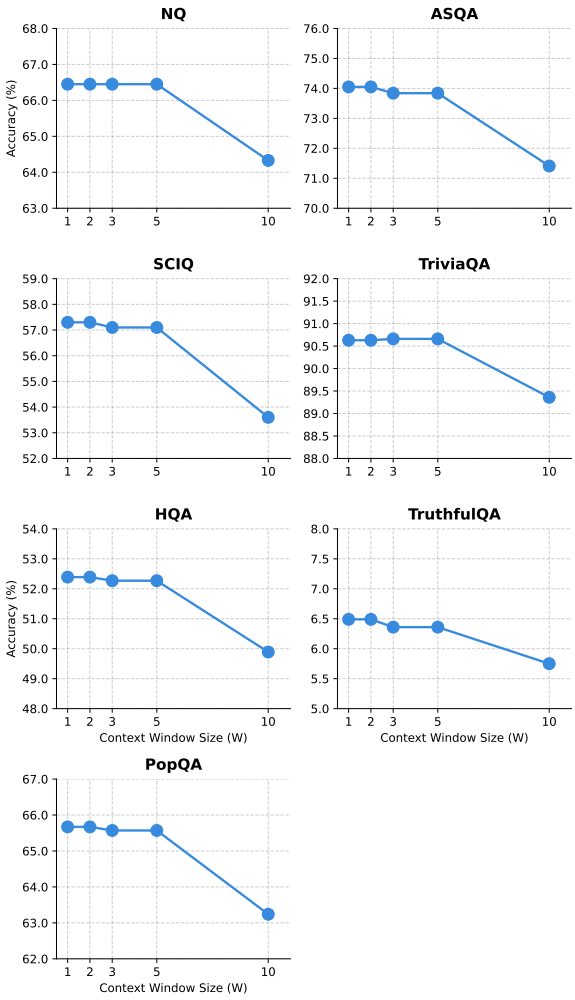

Figure 2: Performance sensitivity to context window size W.

A.10 Effect of Neuron Mining Dataset Size

We additionally analyze the effect of neuron-mining dataset size by selecting neurons using $N \in \{100, 500, 1000\}$ samples from HotpotQA on Llama-3-8B-Instruct. Table 13 reports the performance on NQ, SCIQ, and HotpotQA.

We observe that increasing the number of neuron-mining samples does not necessarily improve downstream performance. In particular, $N = 100$ and $N = 500$ yield comparable results, while performance drops when using $N = 1000$, despite requiring substantially more mining data. The selected neurons also exhibit considerable overlap across different sample sizes. These results suggest that context-aware neurons can be identified with relatively small mining sets.

# Samples	NQ	SCIQ	HQA	Overlap
100	66.44	57.10	52.27	–
500	66.94	57.50	52.63	3/5
1000	64.61	54.10	50.28	3/5

Table 13: Effect of neuron mining dataset size on HotpotQA using LLaMA-3-8B-Instruct. Overlap indicates the percentage of shared neurons compared to the 100-sample setting.

Gold Pos.	Method	# Distractors (N)			
		0	**2**	**4**	**8**
First	RAG	81.0	79.4	77.8	77.4
	SCoNE	84.2	83.6	82.2	81.9
	Δ	+3.2	+4.2	+4.4	**+4.5**
Shuffle	RAG	81.0	79.2	77.4	75.2
	SCoNE	84.2	83.6	81.4	80.5
	Δ	+3.2	+4.4	+4.0	**+5.3**
Last	RAG	81.0	79.8	78.0	75.7
	SCoNE	84.2	82.5	83.1	81.6
	Δ	+3.2	+2.7	+5.1	**+5.9**

Table 14: Controlled noise experiment on 1000 HotpotQA validation examples across different gold-document positions. Each context contains two gold documents and N distractors.

A.11 Validation of Selective Context-Aware Neurons.

We hypothesize that a selective context-aware neuron should respond systematically to the composition of retrieved evidence. As gold documents are progressively replaced by distractors, its activation should also change progressively. Accordingly, the activation under the mixed condition should lie between those under the gold-only and distractor-only conditions. To validate this hypothesis, we analyze the selected neurons on n held-out HotpotQA samples. We use two sample sizes, $n = 1,000$ and $5,000$. For each neuron, we measure the mean activation at the final input position before answer generation under three context settings: GG, containing two gold documents; GD, containing one gold and one distractor; and DD, containing two distractors.

A.12 Controlled Noise Experiments

We evaluate robustness under controlled levels of noise. On the HotpotQA validation split, we construct each context with 2 gold documents and N distractor documents, where $N \in \{0, 2, 4, 8\}$. We keep the same selected neurons and the same 1000 samples across all noise levels, and we vary the gold documents' position: first, shuffle, last. Table 14 shows that vanilla RAG drops steadily as distractors are added. SCoNE degrades more slowly, and its gain grows with noise compared to vanilla RAG. This indicates that SCoNE mitigates performance degradation under accumulating distractors. The effect remains largely invariant to the position of gold documents, suggesting that the improvement is not sensitive to gold-document position.

Neuron	GG	GD	DD	$\lvert GG - DD \rvert$
$n = 1{,}000$				
30@3382	6.848	6.387	5.887	0.961
27@8140	-3.373	-3.433	-3.513	0.140
30@5035	0.188	0.168	0.163	0.025
21@12666	-0.152	-0.122	-0.124	0.028
13@2158	-1.568	-1.415	-1.170	0.398
$n = 5{,}000$				
30@3382	6.870	6.392	5.900	0.970
27@8140	-3.380	-3.430	-3.495	0.115
30@5035	0.187	0.168	0.164	0.024
21@12666	-0.153	-0.123	-0.124	0.029
13@2158	-1.563	-1.412	-1.176	0.387

Table 15: Mean activations of the five selected neurons under gold-only (GG), mixed gold-distractor (GD), and distractor-only (DD) contexts. The notation $l@i$ denotes the i-th intermediate neuron in the l-th FFN layer.

Table 15 shows that across both evaluation sizes, four of the five neurons exhibit a graded activation pattern in which GD lies between GG and DD. This indicates that their activations systematically track the composition of supporting and distracting evidence rather than responding uniformly to retrieved context. The activation patterns and $|\mathrm{GG} - \mathrm{DD}|$ gaps remain nearly unchanged between $n = 1,000$ and $n = 5,000$, indicating that the observed patterns are stable and are not driven by a small evaluation sample. These results suggest that cross-input variability-based mining identifies neurons that selectively respond to the composition of retrieved evidence, supporting their characterization as selective context-aware neurons.

A.13 Evaluation on Out-of-Domain Tasks.

We evaluate SCoNE on three out-of-domain tasks using the lm-evaluation-harness to assess whether neuron enhancement affects performance beyond QA: HellaSwag (Zellers et al., 2019), ARC-Challenge (Clark et al., 2018), and MemoTrap (Liu and Liu, 2023). Table 16 compares SCoNE with vanilla RAG and IRCAN. On HellaSwag and ARC-Challenge, both IRCAN and SCoNE remain close to RAG, with only small changes on both backbones. On MemoTrap, SCoNE improves clearly over both RAG and IRCAN on Llama-3-8B-Instruct.

Method	HellaSwag	ARC-Challenge	MemoTrap
Llama-3-8B-Instruct			
RAG	78.19	62.20	49.15
IRCAN	77.04 (-1.15)	60.58 (-1.62)	58.12 (+8.97)
SCoNE (Ours)	76.55 (-1.64)	59.04 (-3.16)	65.71 (+16.56)
Qwen-2.5-7B-Instruct			
RAG	81.00	65.70	66.56
IRCAN	81.00 (0.00)	66.13 (+0.43)	66.99 (+0.43)
SCoNE (Ours)	80.61 (-0.39)	65.27 (-0.43)	68.06 (+1.50)

Table 16: Accuracy comparison across out-of-domain benchmarks using Llama-3-8B-Instruct (top) and Qwen-2.5-7B-Instruct (bottom).

On Qwen-2.5-7B-Instruct, where RAG already performs strongly, all edited variants remain close to it. Overall, neuron enhancement does not cause severe degradation of general ability, and its effect varies by task rather than uniformly harming out-of-domain performance.

A.14 Experimental Details of Out-of-Domain Tasks

Out-of-Domain results are obtained with the Eluther AI LM Evaluation Harness (Gao et al., 2024). HellaSwag and ARC-Challenge are run 5-shot and scored by length-normalized accuracy (acc_norm), while MemoTrap is run zero-shot and scored by accuracy (acc).